Surya K

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Chennai, India

Profile Summary

A diligent and intellectually curious Electronics and Communication Engineering undergraduate with a strong interest in embedded systems, industrial IoT, robotics, and emerging technologies. Equipped with a solid foundation in electronics, programming, and system-level problem solving, with hands-on experience gained through academic projects, technical competitions, and industry exposure. Possesses a keen eye for detail, a structured approach to problem solving, and the ability to adapt to new technologies and collaborative environments. A team-oriented individual who enjoys taking on technical challenges, learning through practical experimentation, and contributing effectively to team-based projects. Eager to continuously expand technical knowledge and apply engineering skills to develop practical and reliable solutions for real-world problems.

Education

Sri Venkateswara College of Engineering *2023 – 2027*
B.E. Electronics and Communication Engineering ***(CGPA: 8.06/10)*** *Chennai*

D.A.V. Senior Secondary School (Boys), Mogappair *2022 – 2023*
Class XII – All India Senior School Certificate Examination ***(Percentage: 82.4%)*** *Chennai*

D.A.V. Senior Secondary School (Boys), Mogappair *2020 – 2021*
Class X – All India Secondary School Examination ***(Percentage: 86.8%)*** *Chennai*

Experience

OliveIoT *July 2026 – Present*
Embedded Systems and IoT Intern *Chennai, India*

- Contributed to real-time embedded and Industrial IoT applications using STM32, ESP32, and ESP32-S3 platforms, working with UART, SPI, I^2C, GPIO, and FreeRTOS.
- Integrated and tested industrial communication interfaces including RS485, Modbus, LoRa, Ethernet, and GSM, with hands-on involvement in device configuration, communication testing, and debugging.
- Worked on sensor, actuator, and peripheral integration across custom embedded hardware, including environmental sensing, ranging, relay control, biometric interfaces, energy monitoring, and serial peripherals.
- Developed Python-based test utilities and supported hardware bring-up, troubleshooting, and validation of embedded communication and edge-IoT systems across Linux and networked environments.

Syrma SGS Technology Ltd *Dec 2025 – Jan 2026*
Operations Intern *Chennai, India*

- Gained industrial exposure to end-to-end electronics manufacturing across SMT, THT, backend assembly, material flow, inspection, testing, and quality processes.
- Observed PCB assembly and inspection processes including solder paste printing, SPI, pick-and-place, reflow soldering, AOI, wave soldering, ICT/FCT, visual inspection, and rework.
- Gained exposure to RFID manufacturing, cable and wire harnessing, antenna production and RF validation, and EV charger PCBA, with emphasis on ESD safety, process discipline, traceability, and quality assurance.

Projects

ASPLUT – Near-Memory LUT Generator for PIM LLM Inference *July 2026 – Present*

- Architected a serial–parallel near-memory LUT generator for LUT-centric Processing-in-Memory LLM inference, replacing spatially replicated arithmetic with a shared time-multiplexed MAC to materialize 112 half-table entries.
- Established an RTL verification flow using Verilog, Icarus Verilog, and an independent Python golden model, exhaustively validating 28,672 generated LUT entries across all 256 signed activation-pair combinations with zero mismatches.
- Characterized the proposed architecture against an identically constrained baseline, achieving 37.7% lower mapped-netlist area, 51.8% lower total power, 68.3% lower switching power, and a 45.2% improvement in derived maximum frequency.

Raspberry Pi Edge IoT Platform *August – September 2026*

- Configured a Raspberry Pi Zero 2 W as a Linux-based edge-IoT platform, integrating environmental, motion, and gas sensing with GPIO and bus-level peripheral communication.
- Diagnosed and resolved sensor communication issues through I^2C bus scanning, register-level inspection, and Python-based hardware testing, establishing reliable temperature, humidity, and pressure acquisition.
- Integrated an SPI TFT display with a Flask-based local application and enabled embedded media processing and playback using FFmpeg for on-device visualization and interaction.

Motor Health Monitoring System *July 2026*

- Established real-time motor condition monitoring from current, temperature, and vibration inputs, with threshold-based classification of operating states as good, warning, or fault.
- Integrated sensor acquisition and fault logic across ESP32 and STM32 firmware implementations, with UART telemetry and local LED and buzzer indicators for immediate status reporting.
- Connected the monitoring workflow to Firebase for cloud-based data access, extending the system toward predictive maintenance and remote condition monitoring.

Hexapod – Autonomous Multi-Legged Robot *September 2025 – June 2026*

- Architected a two-level robotic control system with STM32 and Zephyr RTOS handling real-time servo control, while a Raspberry Pi 4 and ROS 2 layer support higher-level navigation and autonomous behaviour.
- Derived closed-form inverse kinematics and coordinated tripod, ripple, and wave gait generation for an 18-servo six-legged platform with deterministic 50 Hz motion updates.
- Designed and integrated a custom PCB for power distribution, servo control, and sensor interfacing, and validated locomotion and controller behaviour through ROS 2 and Gazebo simulation.

Roctara – Wall-Climbing Inspection Robot *July 2025 – June 2026*

- Reworked an earlier wall-climbing prototype into a compact, modular inspection platform, replacing propeller-based suction with an EDF-based mechanism to improve adhesion consistency and simplify mechanical iteration.
- Established RF-based wireless control with high-torque N20 gear motors and electronic speed control, while extending the platform with ROS and Gazebo simulation for controller and system validation.
- Trained a YOLO26n-seg crack-segmentation model and coupled it with OpenCV-based skeletonization, distance-transform analysis, and pixel-to-millimeter calibration to estimate crack length, width, and area through a Flask dashboard.

MAKE-A-THON 7.0 & **UPAGRAHA '26** – Event Platforms *February – April 2026*

- Contributed as one of the developers to the official web platforms for Make-A-Thon 7.0 and UPAGRAHA '26, working within a multi-developer student team and using prompt-driven AI-assisted development workflows.
- Built and refined responsive event pages, navigation, timelines, interactive sections, animations, registration information, problem statements, schedules, and sponsor content across both public-facing websites.
- Supported iterative updates to the live platforms and coordinated frontend changes across shared page structures, modular scripts, styles, media assets, and event-specific content.

Technical Skills

Programming: C, C++, Python, MATLAB
Embedded Systems & RTOS: STM32, ESP32, Raspberry Pi, Arduino, FreeRTOS, GPIO
Communication & Networking: UART, SPI, I^2C, RS485, Modbus, LoRa, Ethernet, TCP/IP, HTTP, GSM, Wi-Fi
Robotics, AI & Computer Vision: ROS, ROS 2, Gazebo, RViz, OpenCV, YOLO26, Inverse Kinematics, SLAM
Development Tools & Platforms: Linux, Bash, SSH, Flask, HTML, Antigravity, Codex, Edge Impulse, Firebase, Git, GitHub
Hardware & EDA: KiCad, Proteus, LTspice, AutoCAD, STM32CubeIDE, STM32CubeMX, Arduino IDE

Accomplishments & Certifications

- **High Scorer (Centum) in Engineering Graphics** held by AISSCE 2023.
- **Hackerz'26, SeedStart** – Top 15 among 530+ registrations.
- **Sprinthathon'25** – Top 100 among 3500+ teams in a national-level hackathon.
- **Upagraha'25** – 1st Place among 35 teams in the Blind Chase competition.
- **IMPULSE'26** – Winner, Project Presentation at Chennai Institute of Technology.
- **ZEPHORIA'25** – Winner, Project Expo at VelTech Institute of Science and Technology.
- **Introduction to Industry 4.0 and Industrial Internet of Things** – NPTEL.
- **Electronic Systems Design: Hands-on Circuits and PCB Design with CAD Software** – NPTEL.
- **Electronics Fundamentals** – Etalvis.
- **Matlab and Simulink Onramp** – Mathworks.

Leadership & Campus Involvement

ECEA – Electronics and Communication Engineers' Association, SVCE | 2025 – 2026
Treasurer

- Managed budgeting, fund allocation, and financial coordination for association activities and technical events.
- Supported planning and execution of student-led technical events while coordinating teams and stakeholders.

E-Cell, SVCE | 2024 – 2025
Marketing Associate

- Contributed to event promotion, outreach, and audience engagement activities for entrepreneurship-focused initiatives.

Speakers Forum, SVCE | 2023 – 2024
Event Management Team Member

- Supported event planning, coordination, and on-ground execution of speaker sessions and student activities.